

# Literature-Implied Partial Subcase for arXiv:0902.3422

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## Status

**Status:** literature\_implied\_answer (partial subcase).

This note records a known-literature implication answering only the unconditional-basis subcase of a question from Lopez Perez [1]. It is not a full answer to the unrestricted question.

## Source Question

The source is arXiv:0902.3422, “Banach spaces with many boundedly complete basic sequences failing PCP” by Gines Lopez Perez [1]. Remark 2.7(iii), page 8 of the arXiv PDF, asks whether a Banach space must satisfy the convex point of continuity property (CPCP) whenever every semi-normalized basic sequence has a boundedly complete subsequence.

## Supporting Literature

The supporting source is arXiv:2302.11287, “Fixed point property, unconditional basis and Banach spaces containing no copies of  $c_0$ ” by Cleon S. Barroso [2]. Remark 4.15, page 11 of the arXiv PDF, records that if a Banach space  $X$  has an unconditional basis and fails CPCP, then  $X$  contains a  $c_0$ -sequence; the remark cites Schachermayer [3, Corollary 2.11].

## Identification

Let  $X$  be a Banach space with an unconditional basis and suppose every semi-normalized basic sequence in  $X$  has a boundedly complete subsequence. If  $X$  failed CPCP, the implication recorded in Barroso’s Remark 4.15 would give a basic sequence in  $X$  equivalent to the unit vector basis of  $c_0$ . No subsequence of the  $c_0$  basis is boundedly complete: the scalar choice  $a_n = 1$  has uniformly bounded partial sums in  $c_0$  but the series does not converge in norm. This contradicts the source sequential property. Hence  $X$  satisfies CPCP.

## Scope and Search Notes

This answers only the unconditional-basis subcase of the source question. The unrestricted implication remains open from this packet’s bounded pass: the local run indexes had no exact prior entry for arXiv:0902.3422; local source search found the same source question and nearby CPCP/bounded-completeness material, including Barroso’s Remark 4.15; web searches for exact phrases such as

“boundedly complete subsequence” with “convex point of continuity property” did not reveal a full published answer.

## References

## References

- [1] G. Lopez Perez, *Banach spaces with many boundedly complete basic sequences failing PCP*, arXiv:0902.3422, 2009.
- [2] C. S. Barroso, *Fixed point property, unconditional basis and Banach spaces containing no copies of  $c_0$* , arXiv:2302.11287, 2023.
- [3] W. Schachermayer, *The Radon-Nikodym property and Krein-Milman property are equivalent for strongly regular sets*, Transactions of the American Mathematical Society 303 (1987), 673–687.